Final Project Report

MEE422: Mechanical Engineering Design

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1. PROJECT OVERVIEW

The objective of this project is to design and build an automated conveyor-based waste sorting system capable of separating municipal solid waste into three categories: ferrous metals, non-ferrous metals, and plastics. The system aims to provide a low-cost, safe, and mechanically reliable solution that reduces manual labor and improves recycling efficiency.

The prototype consists of a steel frame supporting a hopper of volume 0.25 m^3, which stores the incoming mixed waste. A flat-belt conveyor transports the material from the hopper toward the sorting section at a controlled speed of 0.2 m/s, ensuring a uniform and continuous flow.

The sorting mechanism combines mechanical motion with an image-processing unit to classify each item in real time. As each object passes beneath the camera, its material type is detected, and the corresponding pusher is actuated to divert it into the correct collection bin. Three bins are positioned at the end of the conveyor to collect:

- Ferrous metals
- Non-ferrous metals
- Plastics

The machine structure is composed of steel profiles, a rigid welded frame, rotating shafts supported by radial bearings, and a motor-driven belt assembly. The driving motor operates at 636.619 rpm and transmits motion to the conveyor through a shaft system, ensuring smooth and continuous belt operation.

Major components of the system include:

- Steel Support Frame: Provides rigidity, stability, and proper elevation for the hopper and conveyor line.
- Hopper (0.25 m^3): Stores mixed waste and funnels the material toward the belt through a controlled outlet.
- Flat-Belt Conveyor: Transports items at constant speed for scanning and sorting.
- Drive Motor and Shafts: A motor with 636.619 rpm drives the belt through rotating shafts supported by bearings.

- Image Processing Unit: Identifies material type using a mounted camera and classification algorithm.
- Pneumatic/Mechanical Pushers: Activated automatically to redirect items into the appropriate bins.
- Collection Bins: Three compartments positioned at the end of the conveyor line for sorted materials.
- Bearings, fasteners, and structural hardware: Ensure smooth rotation, alignment, and assembly integrity.

This system provides a practical and scalable solution for small-scale waste separation. Its modular design makes it suitable for educational use, prototype development, and future enhancement with additional sensing or sorting technologies.

2. DESIGN AND MECHANISM DESCRIPTION

The system operates by receiving mixed waste in the hopper, transporting it on a conveyor belt, detecting the category of each item through image processing, and finally diverting items into the correct collection bin using automated pushers.

1. Feeding Stage – Hopper and Outlet Control

The process begins at the 0.25 m^3 hopper, which stores the input waste. The hopper is supported by a rigid steel frame and designed with sloped walls to prevent material accumulation and promote smooth discharge. The outlet at the bottom delivers items steadily onto the conveyor belt, preventing clumping and ensuring continuous feed.

2. Conveying Stage (Belt Motion and Support)

A flat rubber belt mounted on rotating shafts transports waste items at a constant speed of 0.2 m/s.

- The belt is driven by a 636.619 rpm electric motor, which transfers motion to the main shaft through a pulley or coupling mechanism.
- Radial bearings are used on each shaft end to provide smooth rotation and reduce friction.

- The frame maintains alignment of all components, ensuring stable belt operation and minimal vibration.

This controlled motion ensures that each object passes under the camera one at a time, allowing accurate classification.

3. Detection Stage (Image Processing System)

A camera is positioned above the conveyor belt, synchronized with an image-processing algorithm that identifies the material category based on visual features. The system classifies each object into one of the following categories:

- Ferrous metal
- Non-ferrous metal
- Plastic

Once detected, a control signal activates the corresponding pusher at the appropriate moment, timed according to the conveyor speed.

4. Sorting Stage (Pusher Mechanism)

Three mechanical (or pneumatic) pushers are mounted along the conveyor.
As an item approaches its assigned sorting zone:

- The control system triggers the pusher.
- The pusher extends momentarily, redirecting the object off the belt.
- The object falls into its respective bin positioned directly beneath the pusher.

The pushers are spaced sequentially along the conveyor to separate the three categories clearly:

1. First pusher → ferrous metals
2. Second pusher → non-ferrous metals
3. Third pusher → plastics

Large or unclassified objects continue to the end of the belt as a failsafe.

5. Collection Stage (Waste Bins)

Below the pushers, three collection bins are placed within a dedicated frame. Each bin receives one specific waste category. The layout ensures easy removal, replacement, and inspection.

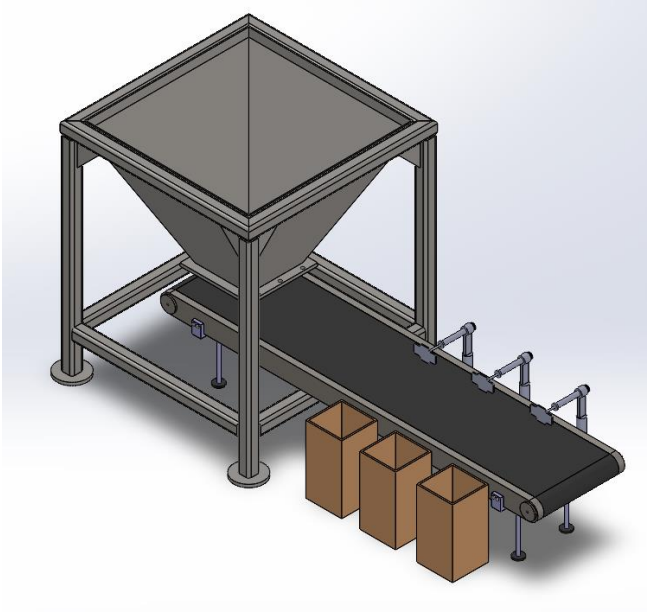


Figure 1 Full Assembly

3.Calculations

1-Bearings

1- Selection of RPM:

Assuming a linear speed of $v = 0.2m/s$

$$V = rw = r\left(\frac{2\pi n}{60}\right) \; thus \; n = v\frac{60}{2\pi r}$$

$$OD_{shaft} = 6mm$$

$$r = 3mm$$

$$n = 636.619 \; RPM$$

3.1 Bearings of the Shafts

Np = 636.619 RPM

For reliability = 90% and L10 = 10^6

W= (mass of the waste+ mass of the whole conveyor belt) x 9.81 = 6 x 9.81= 58.86N

$$Fd = \frac{W}{4} = 14.715N$$

Table 11–4

Bearing-Life Recommendations for Various Classes of Machinery

Type of Application	Life, kh
Instruments and apparatus for infrequent use	Up to 0.5
Aircraft engines	0.5–2
Machines for short or intermittent operation where service interruption is of minor importance	4–8
Machines for intermittent service where reliable operation is of great importance	8–14
Machines for 8-h service that are not always fully utilized	14–20
Machines for 8-h service that are fully utilized	20–30
Machines for continuous 24-h service	50–60
Machines for continuous 24-h service where reliability is of extreme importance	100–200

Figure1 Bearing-Life Recommendations for Various Classes of Machinery Table

Using Table 11-4, Life of our machine is equal to 19 kh.

$$C10 = F_D(l_D n_D 60)^{\frac{1}{a}}$$

$C_{10} = 14.715 \sqrt[3]{\frac{19000x636.619x60}{10^6}} = 132.24\ N$We can select any bore diameter listed in Table 11-2 and still remain within safe operating limits. For our design, we chose a bearing with a 6 mm bore diameter to match the shaft size, and an outer diameter of 20 mm.

Table 11–2

Dimensions and Load Ratings for Single-Row 02-Series Deep-Groove and Angular-Contact Ball Bearings

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder Diameter, mm		Load Ratings, kN			
						Deep Groove		Angular Contact	
				d_S	d_H	C_{10}	C_0	C_{10}	C_0
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6
45	85	19	1.0	52	77	33.2	18.6	35.8	21.2
50	90	20	1.0	56	82	35.1	19.6	37.7	22.8
55	100	21	1.5	63	90	43.6	25.0	46.2	28.5
60	110	22	1.5	70	99	47.5	28.0	55.9	35.5
65	120	23	1.5	74	109	55.9	34.0	63.7	41.5
70	125	24	1.5	79	114	61.8	37.5	68.9	45.5
75	130	25	1.5	86	119	66.3	40.5	71.5	49.0
80	140	26	2.0	93	127	70.2	45.0	80.6	55.0
85	150	28	2.0	99	136	83.2	53.0	90.4	63.0
90	160	30	2.0	104	146	95.6	62.0	106	73.5
95	170	32	2.0	110	156	108	69.5	121	85.0

Figure 3 02-seies bearings specifications

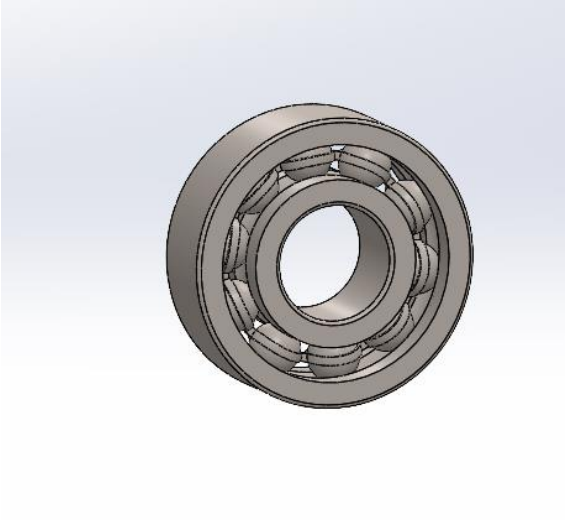


Figure 2 Bearing used in conveyor belt Frame

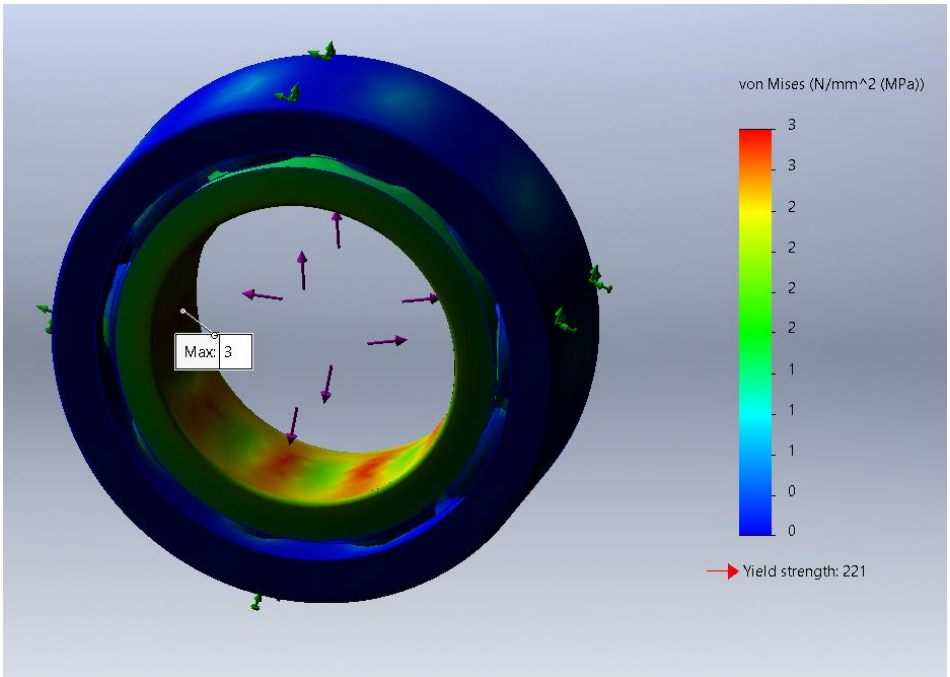


Figure 3 Bearing Stress Analysis

3.2 Shafts for Conveyor Belt

Driving and Driven Shaft:

Considering that the material is **AISI 1045 cold-drawn** steel with a diameter D=6 mm and the motor torque is 1 Nm with an angular speed of 636.619 RPM:

Sut = 585 MPa

Sy = 450 MPa

S'e = 0.5 x 585 = 292.5 MPa. If Sut is less than 1400Mpa

surface condition modification factor:

$$k_a = a\, S_{ut}^{b}$$

Surface Finish	Factor a S_{ut} kpsi	Factor a S_{ut} MPa	Exponent b
Ground	1.34	1.58	−0.085
Machined or cold-drawn	2.70	4.51	−0.265
Hot-rolled	14.4	57.7	−0.718
As-forged	39.9	272.	−0.995

Figure 4 surface condition modification factor.

$$Ka = 4.51 \times 585^{-0.265} = 0.833$$

size modification factor:

$$k_b = \begin{cases} 1.24\, d^{-0.107} & 2.79 \le d \le 51\ mm \\ 1.51\, d^{-0.157} & 51 < d \le 254\ mm \end{cases}$$

Figure 5 size modification factor.

Kb = 1.24$d^{-0.107}$ (knowing that 2.79 < d < 51mm)

Kb = 1.24 x $6^{-0.107} = 1.024$

load modification factor:

$$k_c = \begin{cases} 1 & bending \\ 0.85 & axial \\ 0.59 & torsion \end{cases}$$

Figure 6 Load modification factor.

Kc = 1 (bending and torsion)

reliability factor:

$$k_e = 1 - 0.08\, z_a$$

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

Figure 7 Reliability factor.

Ke = 1 - 0.08 x 1.288 = 0.897

So, the reliability is approximately 90%

miscellaneous-effects modification factor:

Kd = Kf = 1

Se = (Ka x Kb x Kc x Kd x Ke x Kf) x S'e = 223.8 MPa

D/d = 18/6=3

For r = 0.5 mm, r/d = 0.083

Figure A–15–9

Round shaft with shoulder fillet in bending. $\sigma_0 = Mc/I$, where $c = d/2$ and $I = \pi d^4/64$.

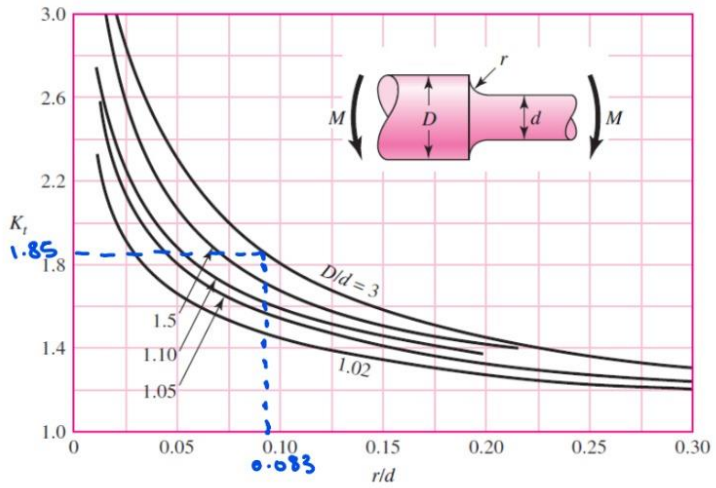


Figure 8 A-15-9 Stress Concentration Due to Bending.

(using figure, A-15-9) → $Kt_{Bending}$ = 1.85

(using figure, A-15-8) → $Kt_s = 1.6$.

now using Figure 6-20 and Figure 6-21 we obtain:

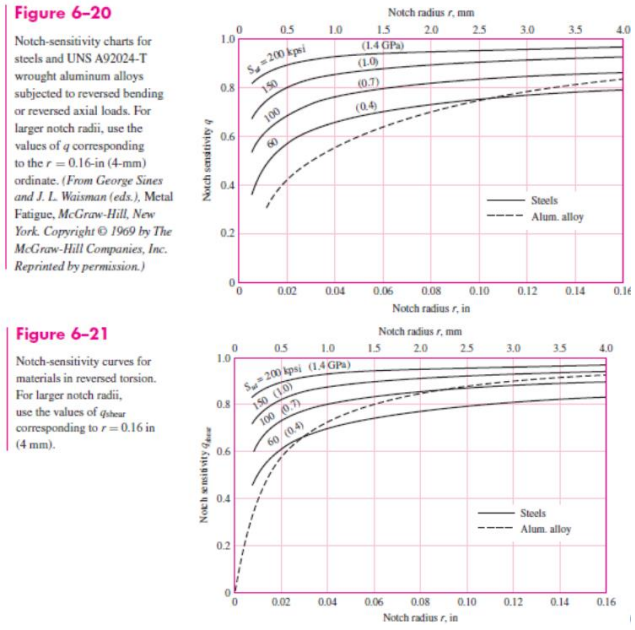


Figure 9 Figures 6-20 and 6-21

$q = 0.62$,

$q_s = 0.7$

$Kf = 1 + q\,(Kt_b - 1) = 1.53$

$Kf_s = 1 + q_s\,(kt_s - 1) = 1.42$

To get the moment on the shoulder Mb:

$W = 6 \times 9.81 = 58.86N$

But we have two shafts so $w_1 = 29.43$ N

- ⇨ $M = 29.43 \times 100 = 2.943$ Nm
- ⇨ $\sigma_b = \frac{My}{I} = 138.8\ MPa = \sigma_{bm}$ $(due\ to\ the\ weight\ being\ constant\ \sigma_a = 0)$

Since the torque on the driving shaft is nearly identical to that on the driven shaft (because losses are very small), we can consider the torque on both shafts to be effectively the same.

- ⇨ $T_a = T_m = 1000$ Nmm

⇨ $\tau_a = \tau_m = \frac{Tr}{J} = \frac{1000*3}{\pi*\frac{6^4}{32}} = 23.58\ MPa$

$$\sigma'_a = \left\{ \left[(K_f)_{\text{bending}} (\sigma_a)_{\text{bending}} + (K_f)_{\text{axial}} \frac{(\sigma_a)_{\text{axial}}}{0.85} \right]^2 + 3 \left[(K_{fs})_{\text{torsion}} (\tau_a)_{\text{torsion}} \right]^2 \right\}^{1/2}$$

$$\sigma'_m = \left\{ \left[(K_f)_{\text{bending}} (\sigma_m)_{\text{bending}} + (K_f)_{\text{axial}} (\sigma_m)_{\text{axial}} \right]^2 + 3 \left[(K_{fs})_{\text{torsion}} (\tau_m)_{\text{torsion}} \right]^2 \right\}^{1/2}$$

$$\sigma'_a = 58\ MPa$$

$$\sigma'_m = 220.14\ MPa$$

Using modified Goodman:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n}$$

n = 1.57 ➔ **infinite life**

Using langer:

$$\sigma_a + \sigma_m = \frac{S_y}{n}$$

n = 1.62 → Infinite life

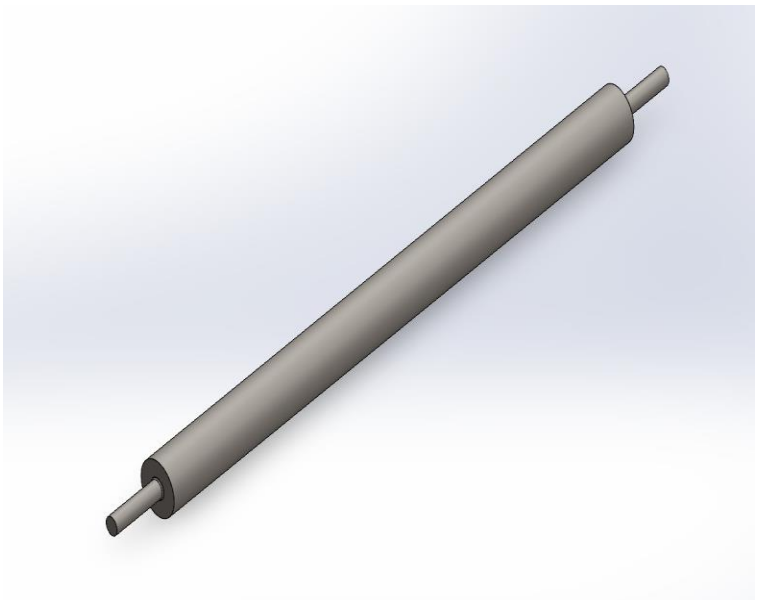


Figure 10 Driven Shaft

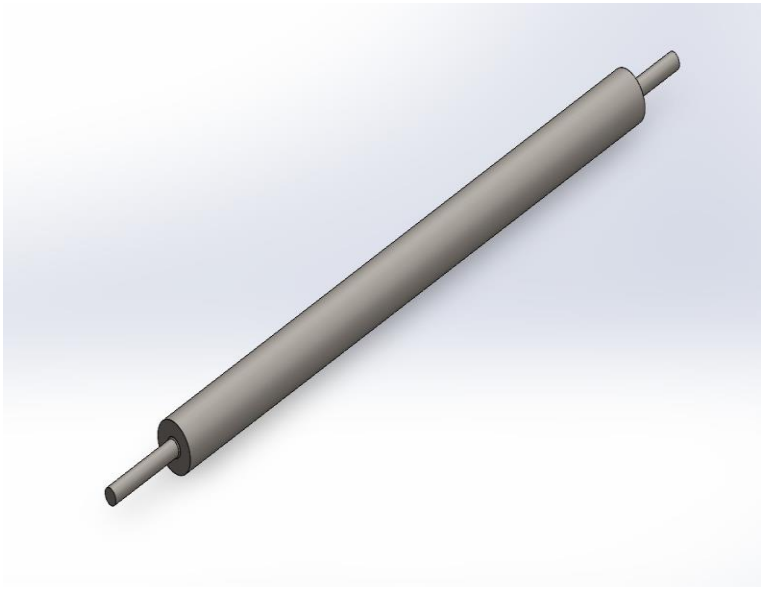


Figure 11 Driving shaft

It's clear that the driving shaft is made longer on the left side (the thinner portion) to provide room for attaching a coupler and the motor.

3.3 Legs of the Conveyor

For the legs, we used stainless steel material with S_y = 172.4 MPa and E = 200 GPa.

The force acting on the legs is the total weight divided by four since there are 4 legs.

Thus, the weight on each leg is $W_L = \frac{30*9.81}{4} = 73.6N$

σ_L = N/A = 73.6/ πr^2 where r= 5mm

⇨ $\sigma_L = 0.94\ MPa$

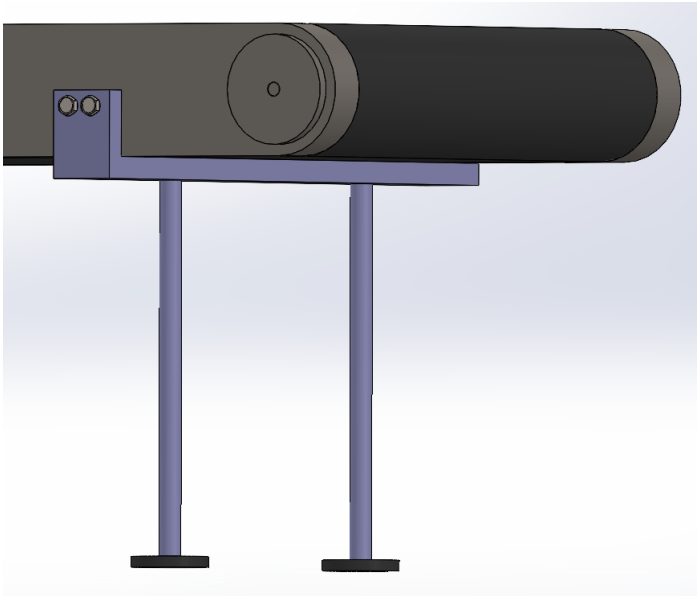


Figure 12 Legs of the Machine

We also added rubber soles underneath each leg so that the machine will stay in place.

We have to check buckling:

$\frac{\pi r\text{^}4}{4} = \frac{P*n*l^2}{\pi^2*E}$ (P_{cr}= 73.6 N, l =170mm, E=200GPa)

$$\frac{P_{cr}}{A} = \frac{C\,\pi^2 E}{(l/k)^2} \qquad P_{cr} = \frac{\pi^2 EI}{l^2}$$

Solving for r, we get r=1.1mm

we have to take a radius greater than 1.1mm

The radius of the leg is r = 5mm > 1.1 mm, **therefore safe from buckling**.

Now we did stress analysis on the frame legs in SolidWorks using Von Misses Criteria. The results are shown below, with max stress being equal to 2 MPa (on the whole part of the legs ≈ 2x σ_L)

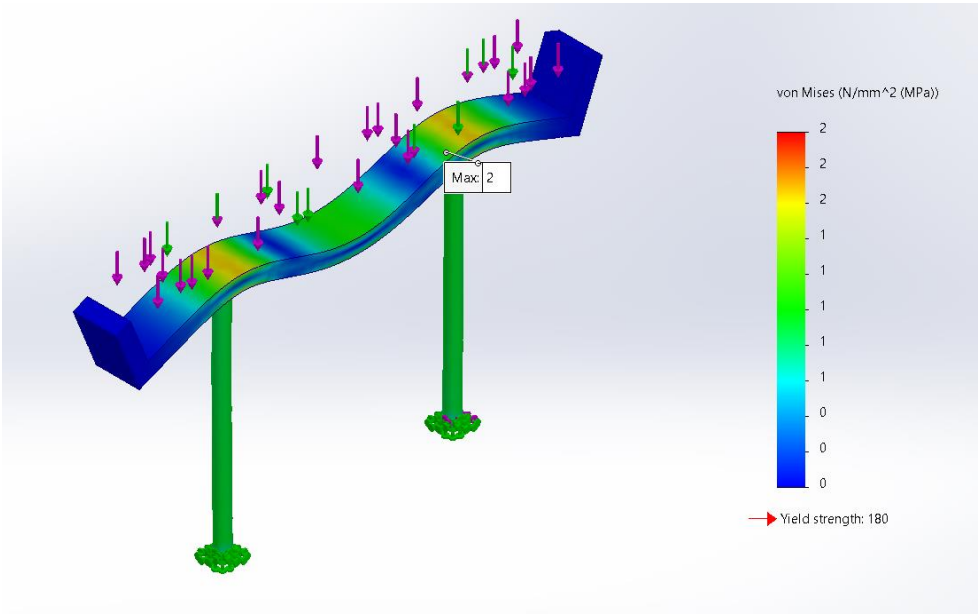


Figure 13 Stress Analysis on Legs

3.4 Bolts of the Frame Legs:

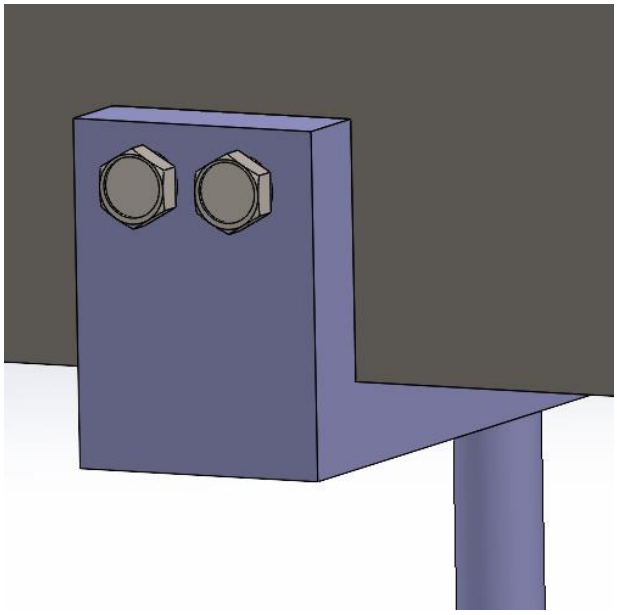


Figure 14 bolts fixing the conveyor belt frame to the legs

Bolts: M4 bolts; Mass = 0.87 grams

Diameter: d = 4 mm

Length: 35 mm

Threaded length: 16mm

Nuts lengths: 4.6 mm

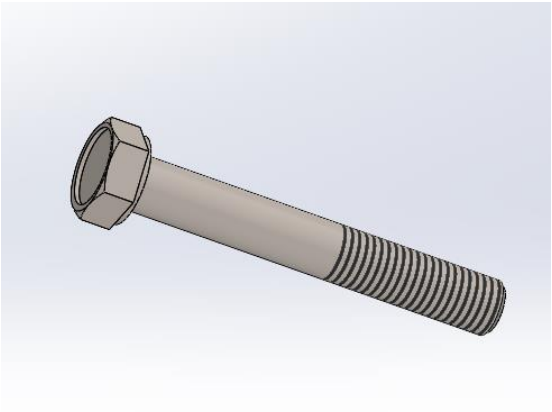


Figure 15 M4 bolts used for connecting the frame to the legs

The thickness of the frame is 20 mm, and the thickness of the legs is 10 mm ➔ l = 30 mm.

$L > l + H$ ➔ $L > 30 + 4.6 = 34.6\ mm$ ➔ we select for the bolt to be 35 mm long.

$LT = 2d + 6mm\ (for\ L \leq 125\ mm, d \leq 48mm)$➔ $LT = 14\ mm$.

$ld = L - LT = 35 - 14 = 21\ mm$.

$lt = l - ld = 30 - 21 = 9\ mm$.

$Ad = \frac{\pi d^2}{4} = \frac{\pi 4^2}{4} = 12.566\ mm^2$

$At = 8.78\ mm^2$ (from Table 8-1).

Then Fastener Stiffness: $k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$

Since we are using an alloy steel screw, the elastic modules E=209.77MPa.

$$\Rightarrow k_b = \frac{(12.566)(8.78)(209.77)}{(12.566)(9)+(8.78)(21)} = 77801.14\ N/mm$$

When determining the dimensions of the hex bolt, we performed calculations for the shear stress it experiences:

Total mass of the assembly above the bolts (assuming there is waste of 4 kg):

M =30 kg (approximately)

Then, the weight on each bolt is $W_b = \frac{30*9.81}{8} = 37\ N$

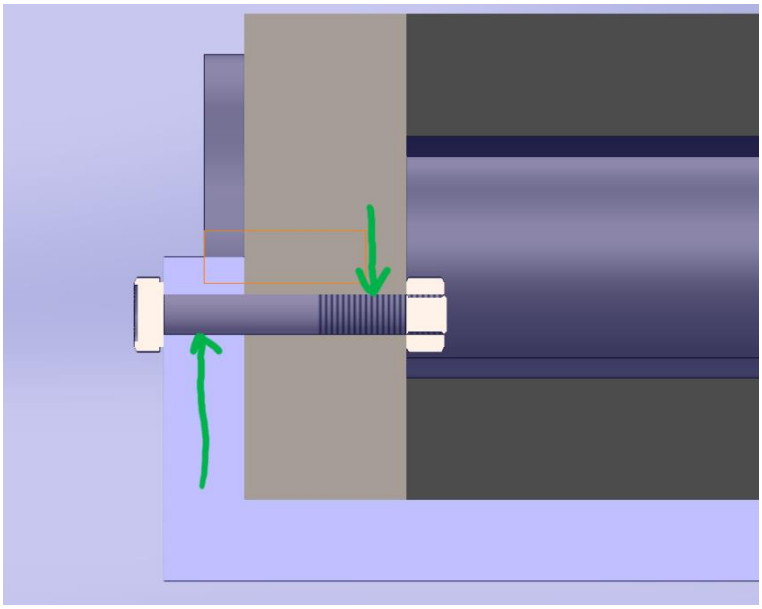


Figure 16 cut showing the forces acting on the bolts

The bolt is subjected to only pure shear, as seen in the figure above which indicates that:

$$\tau = \frac{F}{A} = \frac{F}{\pi * \frac{d^2}{4}} = \frac{37}{\pi * \frac{4^2}{4}} = 2.944\ MPa$$

From the material properties table, 1060 Alloy has a yield strength of 27.5742 N/mm^2

- ⇨ The factor of safety is:

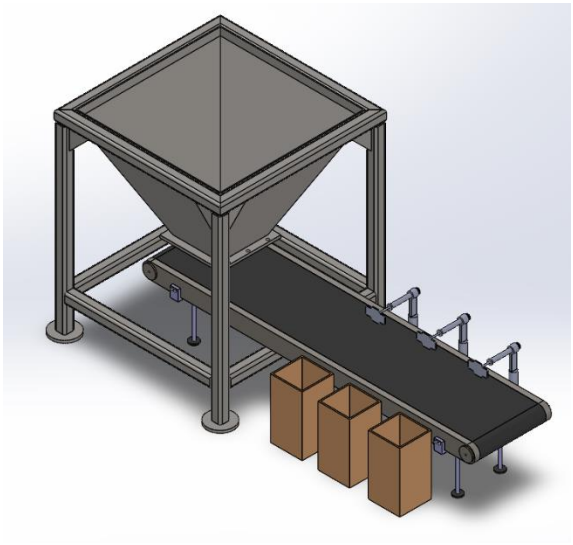
$$n = \frac{Sy}{\tau_{max}} = \frac{27.574}{2.944} = 9.36$$

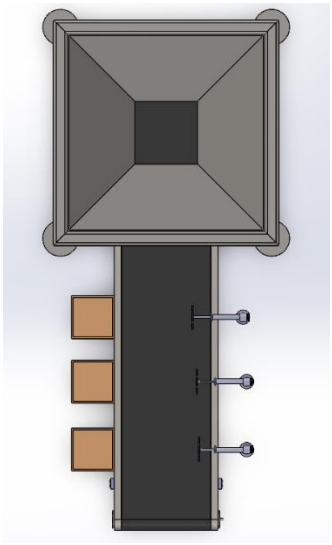
Bill of Materials

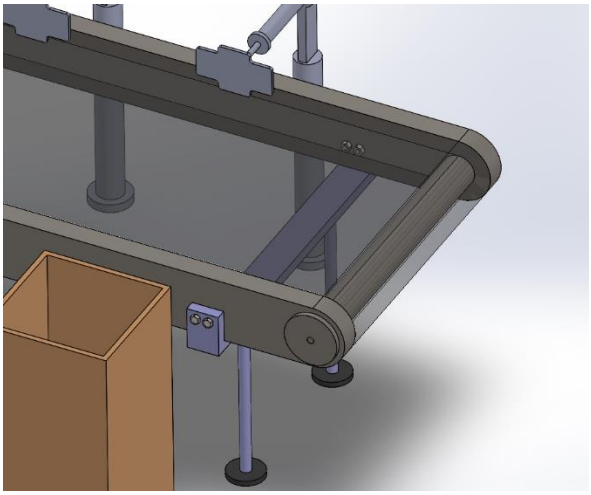
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Belt		1
2	Conveyor frame		1
3	Frame Legs		2
4	B18.2.3.2M - Formed hex screw, M4 x 0.8 x 35 --16WC		8
5	B18.2.4.1M - Hex nut, Style 1, M4 x 0.8 --D-N		8
6	AFBMA 12.1.4.1 - 0060-15 - 10,DE,AC,10_68		4
7	Soles		4
8	Driver Shaft		1
9	Driven Shaft		1
10	Hooper_Frame		1
11	Hopper		1
12	Actuator Stand		3
13	cylinder actuator		3
14	Plunger		3
15	holding Box		3

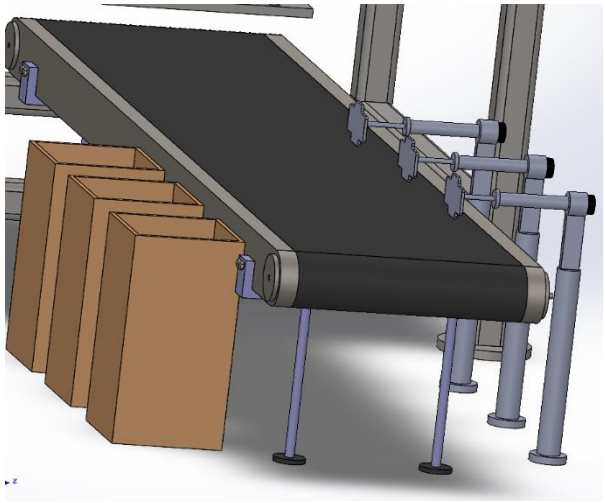
Figure 17 Bill of Materials.

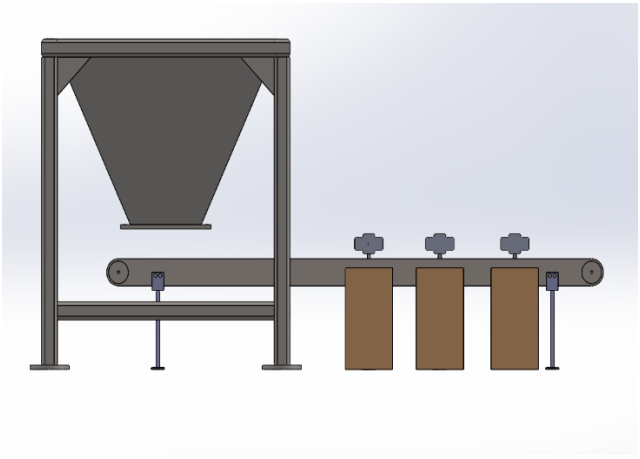
Rendered Images











Shop Drawings

